

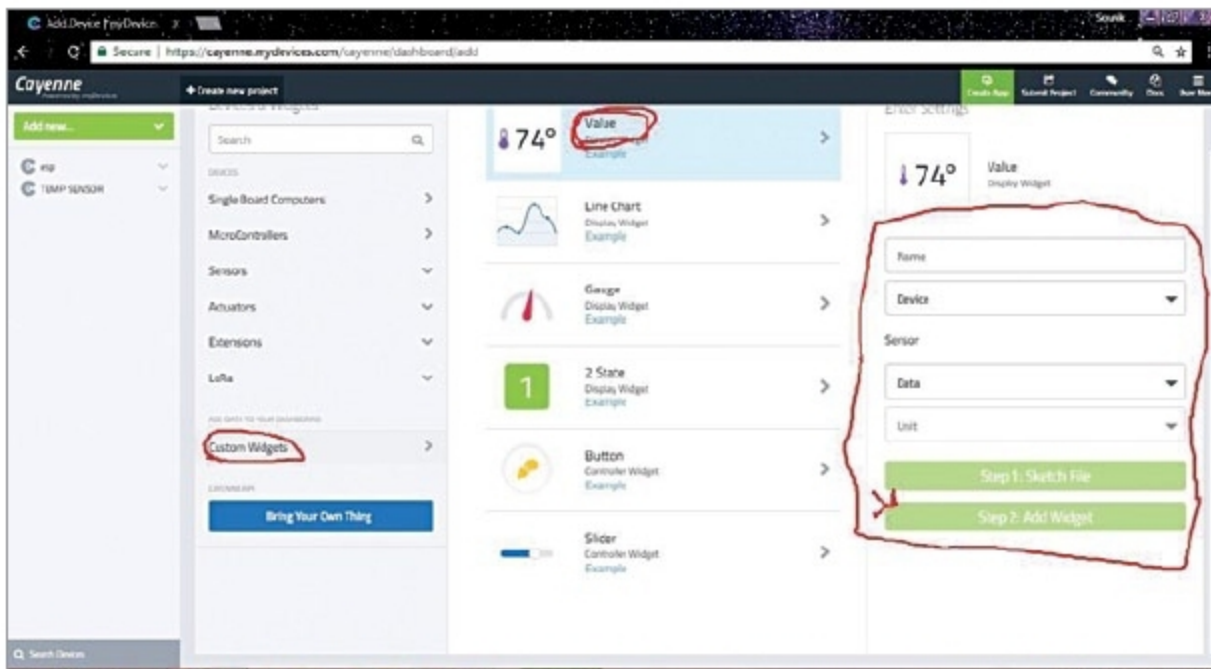


Fig. 8: Custom widgets

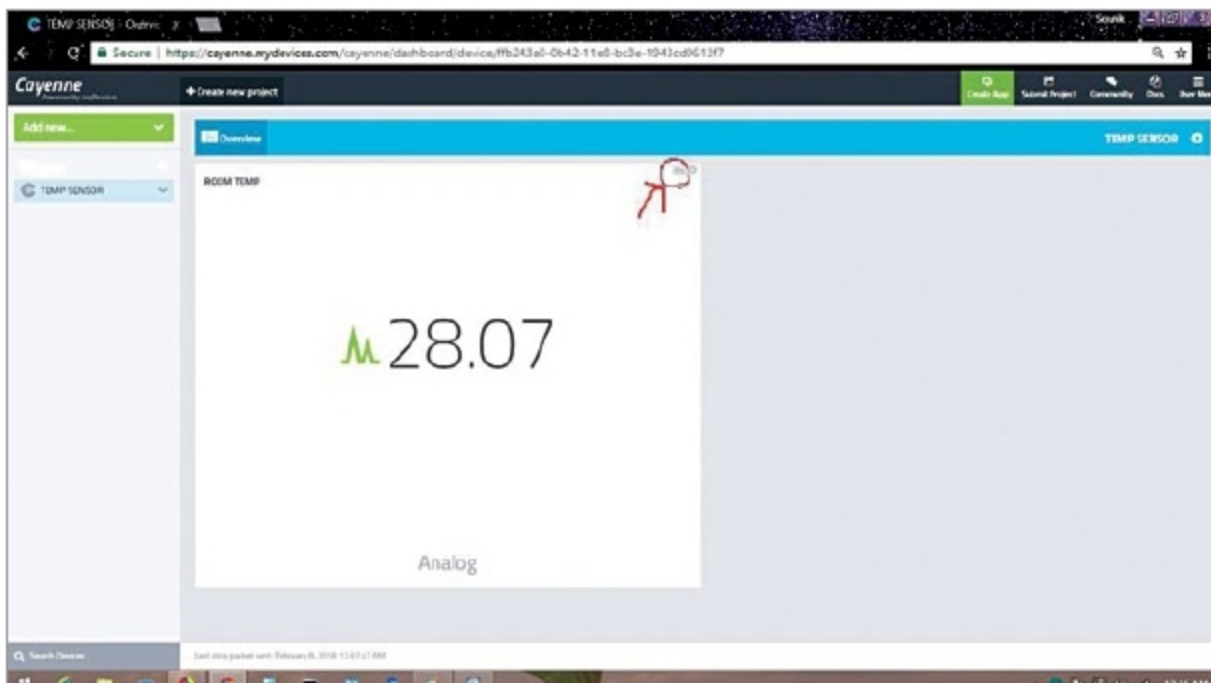


Fig. 9: Temperature data on Cayenne

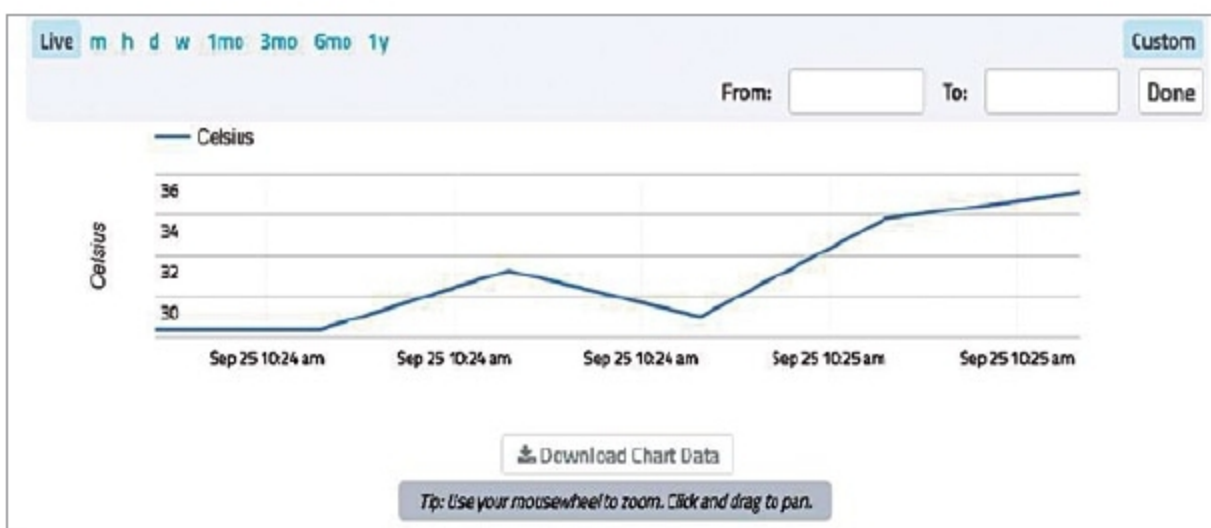


Fig. 10: Graphical representation of temperature data on Cayenne

```
char username[] = "xxxxx";
char password[] = "xxxxx";
char clientID[] = "xxxx";
```

After adding all details in the code, compile it and upload the program (sketch) using a standard USB cable to NodeMCU.

Hardware interfacing with Cayenne IoT platform

To interface the circuit with the IoT platform, open <https://cayenne.mydevices.com> on any browser. Click on Sign Up Free (Fig. 3). Fill all the fields

EFY Note

The source code of this project is available for free download at source.efymag.com

and click on Get Started Free.

Next, click on Bring Your Own Things (Fig. 4) from Cayenne API.

Copy MQTT username, password and client ID from Create App (Fig. 5). Paste these respective details under username, password and client ID in Arduino source code (cayenne_sensor.ino), along with your Wi-Fi name and password.

After successfully compiling and uploading the code to NodeMCU, open Serial Monitor in Arduino IDE. You will see ESP8266 connected to Wi-Fi. After the connection is established, the previous page is automatically updated on Cayenne. A new dashboard opens in the browser. Cayenne generates an ID and a device icon for your device (Fig. 6)—you may change it in Settings.

Click on Add new and then Device/Widget in Settings (Fig. 7).

Click on Custom Widgets and then value, and populate all fields (Fig. 8). The channel number should be 1. (Make sure the channel number is same as in code.) Now, click on Add Widget.

When a connection is made, sensor data gets uploaded to Cayenne. Temperature data on Cayenne is shown in Fig. 9. You can get a graphical representation of temperature data by clicking on the encircled icon shown in Fig. 10.

You can also download Cayenne Android app by myDevices from Google Play Store. The rest of the procedure is the same for accessing temperature data on your smartphone. **EFY**



Souvik Kumar Das is passionate about electronics and MCU-based system design



Shibendu Mahata is M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. He has written in the domain of signal processing in many reputed international journals